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# April 2015

## Team: SC, RM, RS, LC (75%), NW (50%)

## Strategy

- [LC] Research and reporting on landscape of primary mortgage market
    - Concentration of general origination industry and Non-bank lenders
    - Historical trends in average closing costs, cost by State, and summary of different types of closing costs
    - Changes in LLPA costs
    - Issuance of specified pools
    - FHA Single-family application activity
    - History: Guarantee fees, PMI, MIP
- [LC] IOS Monthly Performance Report: Enhanced
- [SC] Quarterly Call Presentation

## Modeling

- **Scale:** Reorganize and refactor existing CCMQuantlib fixed income analytics library (written by Bin Shao) as Scale. See the Scale presentation (distributed separately) for an overarching view of the goals of this library.
    - [JH, NW] New Scale: Work on productionizing Quantlib code that replicates Bin's code. If successful, this approach would provide an excellent platform for our Scale project. Quantlib is:
        - Reasonably well-tested: it's been around since 2000
        - Well-maintained: There are at least a couple of releases every year
        - Rich feature set: Quantlib functionality includes a significant collection of utilities (Date libraries, optimization engines, splines etc.), yield curve builders, interest rate models etc.
        - Documented code
    - [NW] Improved curve construction in Quantlib through judicious selection of interpolation points and convexity adjustments.
    - [NW] Productionized Yield Curve Builder and started saving Spot and 3-month forward rates to time series database on a daily basis
    - [NW] Class structure for Hull-White model
- **Accordion** [RM]: Roll-rate model for GSE loan-level data
    - Automated creation of replines

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- Refined model regressions
- CRT deal forecasts as a function of HPA and Mortgage Rates
- Accordion presentation

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# May 2015 Pipeline

## Team: SC, RM, LC (75%), NW (20%)

## Strategy

- [LC] Research and reporting on landscape of primary mortgage market
    - Loan-level calculation of refinanceable universe using LLPAs and PMI
- [SC] Refinement of IOS Basis Model
- [SC] Research on primary mortgage market

## Modeling

- **Scale** [NW]**:** Implement 1-factor Hull-White model in Scale using Quantlib
- **Accordion** [RM, LC]**:** Roll-rate model for GSE loan-level data
    - Regularly obtain and store CRT monthly files from eMBS
    - Generate a daily forecast of HPA/mortgage rates for the base case.
    - Generate daily delinquency/default vectors for CRT deals using stress scenarios from Moody's/Economy.com and a base case.
    - Generate daily yield numbers for various CRT deals and tranches using these delinquency/default vectors.
    - Create a monthly surveillance report based on the CRT monthly files
- **Baton** [RM]: Agency prepayment-model
    - Curtailment and Default modules